Research Proposal

Is GDP a True Measure of a Nation's Growth and Success?

II. Introduction and Literature Review

A. Identification of the Specific Problem Area

Gross Domestic Product (GDP) is probably the most familiar number used to talk about the performance of a national economy. In simple terms, GDP measures the monetary value of final goods and services produced within an economy during a given period. GDP per capita divides this output by population and is often used when comparing the average level of economic activity across countries. GDP growth then refers to the change in real GDP over time. These measures are useful because they give governments, businesses, and researchers a common way to observe changes in production and economic activity (International Monetary Fund, 2018; World Bank, n.d.-a).

The problem begins when a measure designed to describe economic production is treated as if it were a complete measure of a country's overall success. A country may have a high or rapidly growing GDP while still having large inequalities in income, weak health or education outcomes for some groups, or substantial environmental pressures. GDP also does not directly record unpaid household work, leisure, many aspects of people's quality of life, or how the benefits of economic activity are distributed. The IMF explicitly explains that GDP is not a measure of overall well-being and points to environmental damage, depletion of natural resources, leisure, and distribution as issues that GDP does not capture adequately (International Monetary Fund, 2018).

This research therefore focuses on a specific measurement problem: *how much can GDP, especially GDP per capita, tell us about broader national progress?* The purpose is not to argue that GDP is useless. Economic production clearly matters, and higher income can be associated with improvements in living conditions. The question is whether GDP alone provides enough information to describe growth and success when success includes economic conditions as well as health, education, distribution, and sustainability. This is important because the measures used to describe progress can influence what receives attention in public policy and economic discussion (Stiglitz et al., 2018; OECD, n.d.).

B. Prevalence and Scope of the Problem

The issue is international rather than limited to one country. Countries with similar GDP per capita can have different outcomes in life expectancy, schooling, inequality, and environmental pressure. Output can also rise without every dimension of well-being improving at the same pace. This makes a multidimensional comparison useful.

The Human Development Index (HDI) is one of the best-known alternatives used for this purpose. UNDP created the HDI to emphasize that people and their capabilities should be central when assessing development, rather than economic growth alone. The

HDI combines three broad dimensions: a long and healthy life, knowledge, and a decent standard of living (UNDP, 1990). The current Human Development Data Center provides the corresponding measures used in this study (UNDP, 2025). UNDP also publishes the Inequality-adjusted HDI (IHDI), which accounts for inequality in the distribution of achievements, and the Planetary Pressures-adjusted HDI (PHDI), which adjusts human development for environmental pressures (UNDP, 2025).

This proposal will examine an international panel of countries and territories for the common period 2000–2023. The period is chosen to align the core analysis with the latest UNDP composite-index time series currently available in its Human Development Data Center. The World Bank GDP series extends beyond this period, but using a common window makes comparisons across the main variables more consistent (UNDP, 2025; World Bank, n.d.-a). The final number of countries will depend on matched observations and will be reported transparently after data cleaning. Rather than filling missing values without justification, the study will use the largest defensible matched sample for each analysis.

C. Overview of What We Know: Literature Review

Kuznets's national-income work helped establish systematic measurement of aggregate economic activity and provided an important foundation for modern national accounting (Kuznets, 1934). The strength of GDP-related accounting is comparability: output can be measured regularly and followed over time. Its basic limitation is conceptual. Measuring production does not automatically measure people's quality of life.

Nordhaus and Tobin's *Is Growth Obsolete?* is an early attempt to deal with this distinction. They explored adjustments to conventional economic accounts to produce measures closer to economic welfare (Nordhaus and Tobin, 1972). The work showed the value of including costs and non-market activities, but it also highlighted a difficulty that remains today: broader measures require choices about what should count and how different outcomes should be valued.

The UNDP human-development approach moved the discussion from output alone toward human capabilities and outcomes. The 1990 Human Development Report introduced the HDI to combine health, education, and living standards with income (UNDP, 1990). The current HDI uses life expectancy, education, and income per capita (UNDP, 2025). Its strength is that it makes non-income achievements visible. Its limitation is especially relevant here: because income is already part of HDI, a simple GDP–HDI correlation is partly mechanical. The main analysis will therefore examine health and education separately.

Inequality adds another important dimension. A national average can hide how economic

resources and human-development achievements are distributed. The World Bank's Gini index measures inequality in income or consumption, while UNDP's IHDI adjusts human development for inequality across the population (World Bank, n.d.-b); UNDP's IHDI is described in the Human Development Data Center (UNDP, 2025). These measures allow the study to ask whether countries with similar GDP per capita can have different outcomes once distribution is considered.

Sustainability creates a related problem. GDP records current production but does not by itself show whether that production creates environmental pressures that could affect future well-being. The Stiglitz–Sen–Fitoussi Commission and the OECD therefore argue for broader measurement of well-being, distribution, and sustainability rather than excessive reliance on GDP (Stiglitz, Sen, and Fitoussi, 2009; Stiglitz et al., 2018). UNDP's PHDI is one practical example, adjusting human development for planetary pressure from carbon emissions and material footprint (UNDP, 2025).

There is also an empirical debate about income and subjective well-being. Stevenson and Wolfers reported a positive relationship between GDP per capita and subjective well-being and found that economic growth was associated with rising happiness over time in their data (Stevenson and Wolfers, 2008). Easterlin and colleagues reported a different long-run pattern, finding that happiness did not rise in proportion to income over extended periods (Easterlin et al., 2010). Together, these studies show that the income–well-being relationship depends on the outcome, time period, and empirical design.

Overall, the literature separates economic output, human development, distribution, and sustainability. GDP is useful for production; HDI adds health and education; IHDI adds distribution; and PHDI adds planetary pressure. The remaining question is empirical: how closely do GDP levels and changes line up with these other dimensions in the same international panel?

D. What We Do Not Know: Research Gap

The main gap addressed by this study is the lack of a single, transparent framework in which GDP per capita is compared with several clearly defined dimensions of national progress using the same country-year observations. Much of the existing literature develops alternative measures or debates the conceptual limits of GDP. This proposal takes the next step by making the disagreement measurable.

Three gaps will be examined. First, it is not enough to know that GDP and HDI are correlated because HDI already includes an income dimension. The study will therefore focus on whether GDP per capita is associated with separate health and education outcomes. Second, average GDP can hide distributional differences, so the research will compare GDP with Gini and IHDI measures. Third, a country can achieve high human-

development outcomes while generating relatively high environmental pressures; PHDI provides a way to examine that tension. By bringing these dimensions together, the project can identify where GDP and broader measures tell a similar story and where they tell different stories.

E. Purpose, Research Questions, and Hypotheses

The purpose of this study is to examine whether GDP per capita is a sufficient stand-alone indicator of national growth and success, or whether a broader dashboard provides information that GDP misses.

The research questions are:

1. How strongly is GDP per capita associated with health and education outcomes across countries from 2000 to 2023?
2. Do countries with similar GDP per capita show meaningful differences in inequality-adjusted human development?
3. How often do GDP-per-capita rankings differ from broader development rankings, particularly IHDI and PHDI?
4. Does growth in GDP per capita coincide consistently with improvements in broader measures of human development and sustainability?

The study will test the following hypotheses:

- **H1:** Higher GDP per capita is positively associated with health and education outcomes, but the relationship is not expected to be perfect.
- **H2:** Countries with similar GDP per capita can have substantially different inequality outcomes and inequality-adjusted human-development levels.
- **H3:** GDP-per-capita rankings will not fully reproduce rankings based on broader measures such as IHDI and PHDI.
- **H4:** Increases in GDP per capita will not necessarily be accompanied by equal improvements in measures that account for planetary pressures.

III. Methodology

A. Design

1. Research approach and rationale. This study will use a quantitative research approach because the main questions concern measurable relationships between country-

level indicators. A quantitative design makes it possible to compare many countries and several years instead of relying on a small number of examples.

2. Specific design. The proposed design is a comparative longitudinal panel study using secondary data. Countries will be the units of analysis and the common observation period will be 2000–2023. This design is suitable because it combines cross-country differences with changes over time and allows the study to distinguish a one-year comparison from longer-term patterns.

3. Methods. The study will use descriptive statistics, correlations, rank comparisons, matched-country comparisons, and fixed-effects panel regression. Life expectancy and education will be the main outcomes; IHDI and PHDI will provide broader comparisons. GDP per capita will be the main explanatory variable and will be measured in constant 2015 US dollars (World Bank, n.d.-a). The panel model is:

$$Y_{it} = \alpha + \beta \log(GDPpc_{it}) + \gamma_i + \delta_t + \epsilon_{it},$$

where Y_{it} is a human-development outcome and γ_i and δ_t represent country and year effects. β will be interpreted as an association, not causation.

B. Sample and Procedures

1. Study population and proposed sample. The study population will consist of countries and territories for which comparable World Bank and UNDP data are available. The expected core sample is more than 100 countries from different regions and income levels. The exact number will be reported after the datasets are matched and cleaned.

2. Sampling and procedure. No random sampling will be used. All countries meeting the predefined data criteria will be included. The steps are to download the World Bank and UNDP data, standardize country identifiers, restrict the core panel to 2000–2023, check missing observations, create variable-specific samples, and document exclusions.

3. Informed consent. Informed consent is not required because the study does not recruit or interact with human participants. It uses publicly available, aggregate secondary data published by international organizations. No names, contact details, interviews, surveys, or individual-level records will be collected.

4. Ethical considerations and protection. Although there are no human subjects, research ethics still matter. The study will cite all data sources, preserve the original definitions of variables, avoid fabrication or selective deletion of observations, and record data-download dates and cleaning decisions. Findings will be presented carefully so that a country-level statistic is not interpreted as describing every individual citizen. Missing

data will be reported rather than hidden.

C. Measurement

The main variables and their role in the study are summarized below.

Variable	Measure	Purpose in the study
Economic output	GDP per capita (constant 2015 US$)	Main measure of average economic output; independent variable.
Economic growth	Annual real GDP growth	Captures change in output rather than the level of output.
Health	Life expectancy at birth	Tests whether higher economic output is associated with a concrete human outcome.
Education	Mean and expected years of schooling	Tests whether economic output is reflected in educational achievement.
Human development	HDI	Provides a broad comparison, while recognizing that HDI includes an income dimension.
Distribution	Gini index	Examines whether similar GDP levels are accompanied by different inequality outcomes.
Inequality-adjusted development	IHDI	Examines human-development achievement after accounting for inequality.
Environmental pressure	PHDI	Examines human development after adjustment for planetary pressure.

Table 1: Proposed measurement framework

No survey instrument or interview protocol will be used. The "instrument" for the study is therefore the documented dataset and the reproducible data-processing procedure. For the descriptive ranking analysis, the study will calculate a rank difference:

$$D_i = |Rank(GDPpc_i) - Rank(Y_i)|,$$

where Y_i is a selected broader outcome. A larger value of D_i indicates that a country's position changes more when a different measure of progress is used. This statistic will be used descriptively, not as a universal score of national success.

D. Other Ethical or Political Issues

The topic has political and social relevance because economic indicators are often used to describe national performance. The study will not rank governments or declare one country "best." It will report what each indicator measures and where indicators agree or diverge. "Success" will be treated as multidimensional, and correlations will not be presented as causal effects.

E. Analysis Plan

The analysis will proceed in stages. First, the World Bank and UNDP datasets and metadata will be archived. Second, country-year observations will be cleaned and merged, followed by descriptive statistics and missing-data checks. Third, Pearson and Spearman correlations and scatter plots will examine GDP per capita against health and education outcomes.

Fourth, countries with similar GDP-per-capita levels will be compared on health, education, inequality, IHDI, and PHDI. Fifth, fixed-effects panel models will test whether within-country changes in GDP per capita are associated with changes in health and education after accounting for country and year effects. Robustness checks will include alternative time windows and first-difference models. Results will be reported using tables and graphs.

Interpretation will remain cautious: a strong association would show that GDP contains useful information about an outcome, while a weak association would show that another dimension contains information GDP does not capture well. Neither result alone would make GDP a complete or useless measure.

F. Limitations

The study has several limitations. First, "national success" is not a single objective variable, so the indicators cannot represent every aspect of social progress. Second, country-level data cannot explain differences between individuals or regions. Third, HDI contains an income dimension, which is why health and education components receive greater attention. Fourth, IHDI, Gini, and PHDI have different levels of data coverage. Fifth, the observational design identifies associations rather than causation. Sixth, international datasets are periodically revised and may contain measurement differences. Finally, subjective well-being, unpaid work, safety, political freedoms, and other dimensions are not fully represented.

Week	Activity
1	Finalize research questions, hypotheses, literature review, and variable definitions.
2	Download World Bank and UNDP datasets; archive metadata and source information.
3	Clean country-year data, harmonize identifiers, and document missing observations.
4	Run descriptive statistics, visualizations, correlations, and rank comparisons.
5	Conduct fixed-effects regression and robustness checks.
6	Interpret findings, write the final research report, and document limitations.
7	Prepare the presentation and revise the report using faculty/TA feedback.

Table 2: Proposed project schedule

Project Schedule

IV. References

References

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V. Appendices

V.A. Appendix A: Data Dictionary and Analysis Workflow

Variable	**Source**	**Planned use**
GDP per capita	World Bank, NY.GDP.PCAP.KD	Main economic-output measure.

GDP growth	World Bank	Measure of change in economic output.
Life expectancy	UNDP HDI components	Main health outcome.
Mean/expected years of schooling	UNDP HDI components	Main education outcomes.
HDI	UNDP	Broad human-development comparison.
Gini index	World Bank, SI.POV.GINI	Income/consumption distribution comparison.
IHDI	UNDP	Distribution-adjusted human development.
PHDI	UNDP	Human development adjusted for planetary pressure.

Workflow: collect data → harmonize country-year identifiers → check missingness and definitions → descriptive analysis → correlations → matched-country comparisons → panel regression → robustness checks → interpretation and reporting.

V.B. Appendix B: Consent and Instrument Statement

No consent form is included because the proposed study uses only publicly available, aggregate secondary data and does not involve human participants. No survey, interview, experiment, or participant-level measurement instrument will be used. The appendix is included to make this methodological decision explicit and to address the guideline requirement concerning informed consent and measurement procedures.